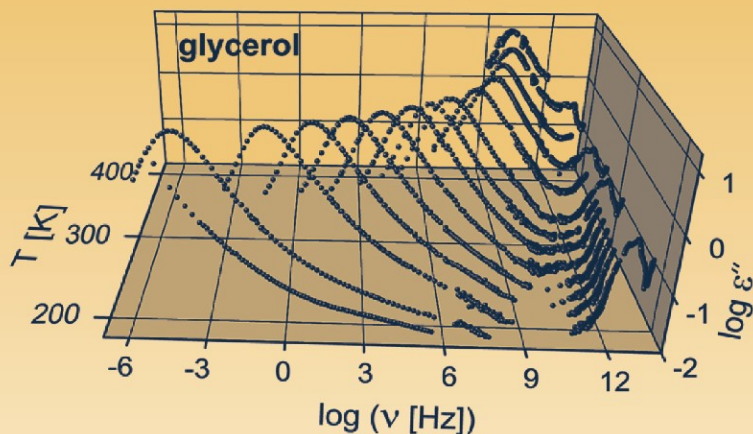


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Editors

Broadband Dielectric Spectroscopy



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Broadband Dielectric Spectroscopy

F. Kremer · A. Schönhals (Eds.)

Broadband Dielectric Spectroscopy

With 393 Figures and 22 Tables



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Foreword

The interaction of electromagnetic (EM) radiation with molecular systems gives rise to quantized transitions between the electronic, vibrational and rotational molecular energy states which may be observed by UV/visible and infra-red absorption spectroscopies at frequencies above about 1 THz (10^{12} Hz). These quantum spectroscopies for molecules in the gaseous, liquid and solid states form a large part of physical chemistry and chemical physics. However, if one asks a fellow scientist “what happens when EM radiation in the range 10^{-6} to 10^{12} Hz is applied to those systems” the answer is usually tentative and incomplete, which shows that a majority of scientists are unfamiliar with the dielectric dispersion and absorption phenomena that occur in this vast frequency range due to (i) dipole relaxation arising from the reorientational motions of molecular dipoles and (ii) electrical conduction arising from the translational motions of electric charges (ions, electrons). This is the domain of *Broadband Dielectric Spectroscopy* (BDS).

At frequencies below about 10^8 Hz a dielectric sample is regarded as a complex electrical impedance $Z^*(\omega)$, expressed in terms of the resistance $R(\omega)$ and capacitance $C(\omega)$, which are frequency-dependent *extensive* properties (here $\omega = 2\pi f/\text{Hz}$ where f is the measuring frequency). The *intensive* complex dielectric quantities of dielectric permittivity $\epsilon^*(\omega)$, electrical modulus $M^*(\omega)$, electrical conductivity $\sigma^*(\omega)$ and resistivity $\rho^*(\omega)$ are immediately derivable from $Z^*(\omega)$. Researchers traditionally use these different quantities to express BDS data for materials, which is a source of confusion when assessing the dielectric/electrical properties of a given material reported in the literature. Dipole relaxation behaviour is normally represented in terms of ϵ^* and electric conduction behaviour in terms of σ^* , Z^* , M^* or ρ^* .

Starting in the late nineteenth century, dielectric measurement techniques were developed for materials such as molecular liquids and solids and moderately-conducting materials such as electrolytes and semiconductors. Transient current methods were used for very low frequencies ($f < 1$ Hz) and a.c. bridges for power, audio, UHF and VHF frequencies (1 to 10^7 Hz). In the 1940s distributed circuit methods were introduced for microwave frequencies (10^8 to 10^{11} Hz) and in the 1970s novel spectroscopic methods were developed for far infrared frequencies (3×10^{11} to 3×10^{12} Hz). In the main, measurements were made point-by-point at each frequency in the range of interest, which was difficult, and time-consuming. This held back the use of dielectric spectroscopy as an investigative physical technique at times when rapid advances were being made in the related techniques of NMR, quasi-elastic light scattering and photon-correlation spectroscopy. Nevertheless extensive dielectric data for a host of organic and inor-

ganic materials were obtained prior to the early 1980s using these methods. Subsequently modern commercial impedance measuring devices for low frequencies and commercial network analysers, time-domain reflectometers and laser spectroscopies for high frequencies became available and were developed for broadband dielectric studies of materials. Automatic measurement and processing methods, made possible by on-line computers used in combination with the new instruments, transformed dielectric measurements of materials. Now it is possible to obtain accurate dielectric data quickly and efficiently over the entire frequency range 10^{-6} to 10^{12} Hz and means that BDS has now taken its rightful place alongside other modern investigative techniques for studying the structure and molecular dynamics of materials.

Debye (1927) established that *dielectric relaxation*, which is the dispersion of the real permittivity (ϵ') and the occurrence of dielectric absorption (ϵ'') in the f -domain for dipolar liquids and solids, was due to the reorientational motions of the molecular dipoles. Many dielectric studies followed, especially those by Smyth (Princeton) and Cole (Brown) that were started in the 1930s. Early areas of study included dipolar liquids (e.g. chlorobenzene, polar solutes in non-polar solvents), rotator-phase crystals (e.g. cyclohexanol, the polymorphs of ice), non-polar polymers (e.g. polyethylene, polypropylene), polar polymers (e.g. polyacrylates, nylons, polyamides). Knowledge of the low frequency permittivity allowed molecular dipole moments to be determined, which was useful for the elucidation of molecular structures prior to the use of modern spectroscopic techniques. The dielectric loss spectra characterized the reorientational dynamics of molecules in the different materials. Also dielectric studies were made for inorganic solids that have ferroelectric properties (e.g. barium titanate) or are semi-conducting (e.g. doped silicon, organic photoconductors and semi-conductors) which have important applications in solid state devices. In parallel, extensive BDS data were obtained for electrolytes, polyelectrolytes, organic and inorganic semi-conductors, giving information on electrical conductivity and hence the mobilities of the effective charge carriers. More recently many BDS studies have been made for novel polymers, glass-forming liquids, liquid crystals (e.g. alkylcyanobiphenyls), polymeric liquid crystals (e.g. polysiloxanes with mesogenic side chains), ferroelectric organic materials (e.g. chiral alkylcyanobiphenyls and their polymeric derivatives), electrolytes (e.g. KCl/H₂O), molten salts (e.g. Ca-K/NO₃) and polyelectrolytes (e.g. polyethylene oxide/salt solutions). Such researches were often motivated by the applications of these materials in devices for modern technology. For example (i) the dielectric anisotropy of the liquid crystal is the source of the optical switching process in liquid crystal displays while (ii) new thin-film solid-state electrolytes are sought for battery applications.

As the data-base for the broadband dielectric behaviour of different materials increased, phenomenological and molecular theories for dipole relaxation and charge conduction were developed. The molecular theories required the deduction of $\epsilon^*(\omega)$ or $\sigma^*(\omega)$ from the field-induced perturbation of the field-free reorientational and translational motions of molecular species. Such theories become extremely difficult for complex motions, e.g. as for multi-site motions of dipoles or ions in crystals or structured liquids. The situation changed completely when Kubo (1957), using linear response theory and time-dependent sta-

tistical mechanics, showed that $\sigma^*(\omega)$ was related, via Fourier transformation, to the field-free mean-squared displacement of charges with time $\langle \Delta R^2(t) \rangle$ or to their velocity correlation function $\langle v(0)v(t) \rangle$. For dielectric relaxation Glarum (1961) and Cole (1965) related $\varepsilon^*(\omega)$, via Fourier transformation, to the field-free dipole moment correlation function $\langle \mu(0)\mu(t) \rangle$ with time. Thus Fourier inversion of $\sigma^*(\omega)$ or $\varepsilon^*(\omega)$ gave direct determinations of these t -dependent molecular properties, which could in turn be fitted by chosen models for motions of molecules or charges. Research texts dealing with dielectric data (mainly for molecular liquids and solids) were published by Smyth (1955), Fröhlich (1958), Hill, Vaughan, Price and Davies (1969) and Böttcher and Bordewijk (1978) and for polymers by McCrum, Read and Williams (1967) and Runt and Fitzgerald (eds) (1997). Obviously the wide range of materials studied in different frequency bands for their dipole relaxation and conduction behaviour, the differing backgrounds of the researchers (physicists, chemists, materials scientists, electrical engineers, chemical engineers, theorists) and the publication of the researches in different journals of physics, physical chemistry/chemical physics, electrical engineering, polymer science and materials science made it increasingly difficult for scientists to monitor overall activities in dielectric spectroscopy. A great help to this effect have been the *International Discussion Meetings on Relaxations in Complex Systems* organized by K. L. Ngai and his associates held in Crete (1990), Alicante (1993) and Vigo (1997), published as special issues of *J Non-Crystalline Solids*, and that held in Crete (2001) where researchers in different areas of dielectrics science have been brought together along with a host of researchers that use related relaxation, scattering and spectroscopic techniques for the study of the dynamics of materials.

The stage has been reached where the foundations of broadband dielectric spectroscopy are well-established in terms of the large body of literature for the dielectric behaviour of dipolar materials and moderately-conducting electrolyte systems, phenomenological and molecular theories that relate $\varepsilon^*(\omega)$ and $\sigma^*(\omega)$ to empirical relaxation functions and to time-dependent molecular properties that give information on the reorientational and translational motions of molecules and charges in molecular liquids and solids. In this book the editors Friedrich Kremer and Andreas Schönhals have built on these foundations, through their own contributions and those of other leading scientists, taking the subject forward into those areas where BDS is making vital and new contributions to our understanding of the dynamics of complex systems. After a summary of the essentials of modern experimental techniques and dielectric theories (Chaps. 1–3) experimental data are shown over the entire frequency range for glasses, supercooled liquids, amorphous polymers (Chaps. 4, 5, 7), (polymeric) liquid crystals (Chap. 10) and semi-conducting disordered materials (Chap. 12) where multiple dipole relaxations are observed and are analysed in terms of particular motional processes. BDS provides a powerful method for studying the dynamics of molecules in confined spaces down to the mesoscopic and molecular levels (Chap. 6). Since the capacitance of a dielectric sample is inversely proportional to its thickness, BDS is highly suitable for studies of ultra-thin films, in contrast to NMR, light scattering and other spectroscopic techniques. The effects of film thickness on molecular dynamics in ultra-thin poly-

mer films are clearly demonstrated (Chap. 11) and provide severe tests of results for glass-forming materials obtained from other physical techniques. Inhomogeneous media (Chap. 13) give rise to interfacial polarization in addition to dipole relaxation and charge-conduction and this is particularly important in multi-phase liquids and polymers, emulsions and biological systems. The component structures of relaxation processes and the questions of dynamic heterogeneity in organic materials are further elucidated by the new, sophisticated techniques of pulsed and non-resonant dielectric hole-burning (Chap. 14). BDS studies provide evidence for the structure and molecular dynamics in all these systems, evidence is obtained on the molecular dynamics which complements that obtained using the related techniques of solvation dynamics (Chap. 15), dynamical mechanical spectroscopy (Chap. 16), multi-dimensional multi-nuclear NMR (Chap. 17) and neutron scattering (Chap. 18). While dielectric relaxation behaviour is usually studied over wide ranges of frequency and sample temperature, the relaxation strength, relaxation time and its distribution are all affected significantly by applied pressure. This 'forgotten variable' (G. Floudas, Chap. 8) can be used to separate overlapping relaxations or induce crystallization – which transforms the relaxation behaviour in crystallizable polymers. Modern techniques allow broadband dielectric measurements to be made in real time for polymerizing systems where a liquid mixture transforms to a glass or an elastomer during reaction (Chap. 9). These studies monitor changes in molecular dynamics during the polymerization reaction and demonstrate the role of diffusion-control on reaction rate when a glass is formed.

Thus the editors and contributors show in this remarkable book that modern BDS techniques, as applied to a wide range of amorphous, crystalline and liquid crystalline systems, can give new and vital information on the reorientational and translational motions of dipolar molecules or electric charges and how the characteristics of these motions vary with temperature and pressure and with the physical condition of the material (bulk or thin film, confined geometries, macroscopic orientation (e.g. for liquid crystals)). They show how the wide frequency range of BDS may be utilised to obtain a detailed knowledge of individual motional processes whose time-scale may be in a range from picoseconds to kiloseconds and show further how this information complements that obtained for the same materials using related relaxation, scattering and spectroscopic techniques. A new researcher, or one from a related field of study, will find this 'state-of-the-art' account of broadband dielectric spectroscopy to be invaluable since *inter alia* it provides clear examples of the power of the technique to elucidate the dynamics of condensed systems, many of which have applied interest. Most important of all, the information provided by the BDS researches described here will stimulate new lines of research, not only into the applications of modern dielectric techniques to new materials and to time-varying systems but also to the development of further novel techniques that will test and extend the conclusions reached presently from BDS, NMR, mechanical relaxation, light scattering, neutron scattering, optical relaxation and related techniques concerning the detailed nature of molecular dynamics in organic and inorganic materials.

Preface

The interaction of electromagnetic waves with matter in the frequency regime between 10^{-6} and 10^{12} Hz is the domain of broadband dielectric spectroscopy. In this extraordinarily extended dynamic range, molecular and collective dipolar fluctuations, charge transport and polarization effects at inner and outer boundaries take place, and determine the dielectric properties of the material under study. Hence broadband dielectric spectroscopy enables us to gain a wealth of information on the dynamics of bound (dipoles) and mobile charge carriers depending on the details of the molecular system. In recent years novel dielectric instrumentation has been developed which allows for automatic measurements in nearly the entire range from ultra low frequencies up to the Far Infra Red.

It is intended that this book be more than a monograph at the leading edge of research. Therefore in three introductory chapters written in the style of a textbook, broadband dielectric spectroscopy is described in its theoretical foundation, its experimental techniques, and in the way dielectric spectra have to be analyzed. In the chapters 4–13, examples are described where the dielectric method has made important contributions to modern scientific topics. This is, of course, far from being a comprehensive overview and corresponds to the research interests of the editors. In chapters 14 and 15, two novel experimental techniques are introduced which are closely related to dielectric spectroscopy. Special attention is given in chapters 16–18 to the comparison between dielectric and other spectroscopic techniques such as mechanical, NMR, and neutron scattering.

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Contents

1	Theory of Dielectric Relaxation	1
	<i>A. Schönhals, F. Kremer</i>	
1.1	Introduction	1
1.2	Electrostatics	3
1.3	Dielectric Relaxation (Dielectric Retardation)	10
1.3.1	Linear Response Theory and Fluctuation Dissipation Theorem	10
1.3.2	Theoretical Considerations (Models)	15
1.3.2.1	Debye Relaxation	17
1.3.2.2	Models for Non-Debye Relaxation	19
	Appendix 1.1 Linear Response Theory	24
	Appendix 1.2 Derivation of the Kramers/Kronig-Relationships	26
	Appendix 1.3 Fluctuation Dissipation Theorem	29
	References	32
2	Broadband Dielectric Measurement Techniques (10^{-6} Hz to 10^{12} Hz)	 35
	<i>F. Kremer, A. Schönhals</i>	
2.1	Introduction	35
2.2	Measurements in the Frequency Domain from 10^{-6} Hz to 10^{11} Hz	37
2.2.1	Fourier Correlation Analysis	37
2.2.2	Dielectric Converters in Combination with Fourier Correlation Analysis	39
2.2.3	Impedance Analysis	41
2.2.4	Integrated Dielectric Analyzers	42
2.2.5	Samples and Sample Cells for Frequencies < 10 MHz	43
2.2.6	RF Reflectometry	43
2.2.7	Network Analysis	45
2.2.8	Impedance and Frequency Ranges Overview	46
2.3	Measurement Systems in the Time Domain from 10^{-6} Hz to 10^{10} Hz	48
2.4	Quasi-optical Set-Ups in the Frequency Range from 10^{10} Hz to 10^{12} Hz	51
2.5	Conclusions	54
	References	56

3	Analysis of Dielectric Spectra	59
	<i>A. Schönhals, F. Kremer</i>	
3.1	Introduction	59
3.2	Dipolar Fluctuations	60
3.2.1	Analysis of Dielectric Spectra – Model Functions	61
3.2.2	Analysis of Dielectric Spectra – Model Free	72
3.3	Fluctuations of Mobile Charge Carriers	81
3.4	Separation of Charges	87
3.4.1	Mesoscopic Scale: Maxwell/Wagner/Sillars Polarization	87
3.4.2	Electrode Polarization	91
3.5	Conclusion – How to Analyze Dielectric Spectra	93
	Appendix 3.1 Fit parameters	94
	References	96
4	The Scaling of the Dynamics of Glasses and Supercooled Liquids	99
	<i>F. Kremer, A. Schönhals</i>	
4.1	Introduction	99
4.2	Theories Describing the Scaling of the Dynamics in Glass-Forming Systems	101
4.3	Temperature Dependence of the Dynamic Glass Transition	108
4.4	Scaling of the Relaxation Time Distribution Function of the Dynamic Glass Transition	116
4.5	Scaling of the Dynamic Glass Transition and the Slow β -Relaxation in Poly(ethylene terephthalate) (PET)	120
4.6	Conclusions	125
	References	127
5	Glassy Dynamics Beyond the α-Relaxation	131
	<i>P. Lunkenheimer, A. Loidl</i>	
5.1	Introduction	131
5.2	Dynamic Processes in Glass-Forming Materials	133
5.3	Broadband Dielectric Spectra of Glass-Forming Materials	136
5.4	The Excess Wing	142
5.4.1	Scaling and Divergent Susceptibility	143
5.4.2	The Excess Wing – a β -Relaxation?	146
5.5	The High-Frequency Response	149
5.5.1	The $\epsilon''(\nu)$ -Minimum	149
5.5.1.1	Phenomenological Description	150
5.5.1.2	Mode-Coupling Theory	152
5.5.1.3	Other Models	157
5.5.2	Comparison with Light and Neutron Scattering	158

5.5.3	The Boson Peak	161
5.6	Conclusions	162
	References	165
6	Molecular Dynamics in Confining Space	171
	<i>F. Kremer, A. Huwe, A. Schönhals, S.A. Rózański</i>	
6.1	Introduction	171
6.2	H-Bond Forming and van-der-Waals Liquids in Zeolitic and Nanoporous Media	172
6.2.1	Glycols in Zeolites and Nanoporous Sol-Gel Glasses	172
6.2.2	Salol in (Lubricated) Nanoporous Sol-Gel Glasses	188
6.3	Polymers in Zeolitic and Nanoporous Media	199
6.3.1	Poly(vinylether) in Mesoporous MCM	199
6.3.2	Poly(propylene glycol) in Nanoporous Sol-Gel Glasses	202
6.3.3	Poly(dimethyl siloxane) in Nanoporous Sol-Gel Glasses	207
6.4	Liquid Crystals in Confining Space	210
6.4.1	Nematic Liquid Crystals in Mesoporous Membranes	210
6.4.2	Ferroelectric Liquid Crystals in Mesoporous Membranes	216
6.5	Conclusions	220
	References	222
7	Molecular Dynamics in Polymer Model Systems	225
	<i>A. Schönhals</i>	
7.1	Introduction	225
7.2	Theoretical Considerations	227
7.2.1	Conformations of Polymeric Chains	227
7.2.2	Dipole Moments of Polymeric Chains	230
7.2.3	Motional Processes in Isolated Chains	233
7.2.3.1	Segmental Motion	233
7.2.3.2	Chain Motion	234
7.3	Dielectric Properties of Polymers in Solution	236
7.3.1	Dilute Solutions	236
7.3.1.1	Segmental Motion	236
7.3.1.2	Chain Motion	238
7.3.2	Concentrated Solutions	238
7.3.2.1	Segmental Motion	238
7.3.2.2	Chain Motion	239
7.3.3	Rod-Like Systems	240
7.4	Dielectric Properties of Polymeric Systems in the Bulk State	241
7.4.1	Amorphous Polymers	241
7.4.1.1	β -Relaxation	242
7.4.1.2	Dynamic Glass Transition (α -Relaxation)	244
7.4.1.3	Relationship Between the β - and the α -Relaxation – the ($\alpha\beta$)-Process	251

7.4.1.4	Normal Mode Relaxation	254
7.4.2	Semicrystalline Polymers	261
7.4.2.1	Morphology of Semicrystalline Polymers	263
7.4.2.2	Polymers with a Medium Degree of Crystallinity	264
7.4.2.3	Polymers with a High Degree of Crystallinity – Polyethylene	271
7.4.3	Polymer Blends	272
7.4.3.1	General Considerations	273
7.4.3.2	Miscible Blends – Concentration Fluctuations	273
7.4.3.3	Immiscible Blends	276
7.4.4	Novel Polymeric Architectures	277
7.4.4.1	Rings	277
7.4.4.2	Stars and Block Copolymers	277
7.4.4.3	Dendrimers	282
7.4.4.4	Networks	283
7.5	Conclusion	283
	Appendix	284
	References	286
8	Effect of Pressure on the Dielectric Spectra of Polymeric Systems	295
	<i>G. Floudas</i>	
8.1	Introduction	295
8.2	Theoretical Background	296
8.2.1	Transition State Theory	296
8.2.2	Models Based on Free Volume Theories	296
8.2.3	Model of Constant Free Volume Compressibility	298
8.2.4	Models of Variable Free Volume Compressibility	299
8.2.5	Problems Related to Free Volume Theories	300
8.2.6	Scaling of the α -Process: an Example	301
8.3	Effect of Pressure on the Separation of the α - and β -Processes	302
8.4	Effect of Pressure on the Local and Global (Chain) Dynamics	305
8.5	Effect of Pressure on the Miscibility of Blends/Copolymers	315
8.5.1	The PI- <i>b</i> -PVE Athermal Diblock Copolymer	317
8.5.2	The PS/PVME Homogeneous Blend	321
8.6	Effect of Pressure on Polymer Crystallization	326
8.6.1	Crystallization Kinetics	338
8.7	Conclusion	344
	References	346
9	Dielectric Spectroscopy of Reactive Network-Forming Polymers 349	
	<i>J. Mijovic</i>	
9.1	Introduction	349
9.2	Polarization Mechanisms	350
9.3	Experimental	351

9.3.1	Principles of Measurements	351
9.3.2	Instrumentation	351
9.3.3	Cells and Sensors	351
9.3.4	In Situ Real Time Monitoring of Cure	352
9.4	Polarization Due to Migrating Charges	353
9.4.1	Modeling Concepts – Equivalent Circuits	353
9.4.2	Graphical Representation and Evaluation of Characteristic Parameters	356
9.4.3	Examples of In Situ Dielectric Cure Monitoring	358
9.5	Polarization by Dipole Orientation	360
9.5.1	Dipolar Relaxation Processes – Classification Scheme	360
9.5.2	Changes in Relaxation Time (τ) During Network Formation	363
9.5.3	Correlation Between Chemical Kinetics and Molecular Dynamics During Network Formation	366
9.5.4	Changes in Relaxation Strength During Network Formation	372
9.5.5	Changes in Relaxation Spectrum During Network Formation	375
9.6	Conclusions	379
	References	381
10	Molecular and Collective Dynamics of (Polymeric) Liquid Crystals	385
	<i>F. Kremer, A. Schönhals</i>	
10.1	Introduction	385
10.2	Molecular and Mesomorphic Structures of Liquid Crystals	385
10.2.1	Low Molecular Weight Systems	385
10.2.2	Polymeric Systems	388
10.3	Theoretical Considerations	388
10.4	Molecular Dynamics	392
10.4.1	Low Molecular Weight Systems	392
10.4.2	Polymeric Systems	396
10.4.2.1	Side Chain Polymers	396
10.4.2.2	Main Chain Polymers	411
10.5	Collective Dynamics in Ferroelectric Liquid Crystalline Systems	412
10.5.1	Low Molecular Weight Systems	412
10.5.2	Polymeric Systems	422
10.6	Conclusions	428
	References	430
11	Molecular Dynamics in Thin Polymer Films	433
	<i>L. Hartmann, K. Fukao, F. Kremer</i>	
11.1	Introduction	433
11.2	Preparation and Methodical Aspects	434
11.2.1	Preparation of Thin Polymer Films	434
11.2.2	Thermal Expansion Spectroscopy	436

11.3	Thin Films of Glass Forming Polymers	438
11.3.1	Poly(methyl methacrylate) (PMMA)	438
11.3.2	Atactic Polystyrene (a-PS)	452
11.4	Thin Films of Grafted Polymers: Poly(γ -benzyl-L-glutamate) (PBLG)	462
11.5	Conclusions	469
	References	472
 12	 The Dielectric Properties of Semiconducting Disordered Materials	 475
	<i>F. Kremer, S. A. Róžański</i>	
12.1	Introduction	475
12.2	Similarities in the a. c. Conductivity of Semiconducting Disordered Materials	480
12.3	Relaxation and Charge Transport in Mixtures of Zwitterionic Polymers and Inorganic Salts	483
12.4	Conclusions	492
	References	493
 13	 Dielectric Properties of Inhomogeneous Media	 495
	<i>P. A. M. Steeman, J. van Turnhout</i>	
13.1	Introduction	495
13.2	Theory	496
13.2.1	Two-Phase Heterogeneous Materials	496
13.2.2	Heterogeneous Materials Containing More than Two Phases	501
13.2.3	Heterogeneous Materials with an Interfacial Layer	502
13.3	Experimental Results	503
13.3.1	Polymeric Blends	503
13.3.2	Filled Polymers	508
13.3.3	Semi-Crystalline Polymers	512
13.3.4	Effect of Processing on Material Homogeneity	516
13.4	Conclusions	519
	References	520
 14	 Principles and Applications of Pulsed Dielectric Spectroscopy and Nonresonant Dielectric Hole Burning	 523
	<i>R. Böhmer, G. Diezemann</i>	
14.1	Introduction	523
14.2	Theoretical Concepts	525
14.2.1	Macroscopic Linear Response	525
14.2.2	Microscopic Reorientation Models	527

14.2.3	Stochastic Dynamics in Moderately Large Electric Fields	533
14.2.3.1	The Model of Asymmetric Double Well Potentials	534
14.2.3.2	The Box Model	540
14.3	Experimental Aspects	542
14.4	Applications	545
14.4.1	Supercooled Liquids and Plastic Crystals	545
14.4.1.1	Pulsed Dielectric Spectroscopy	545
14.4.1.2	Nonresonant Spectral Hole Burning	548
14.4.2	Ion Conducting Glasses	552
14.4.3	Crystals with Frozen-In Disorder	556
14.5	Related Experimental Methods	559
14.6	Conclusions	562
	Appendix 14.1	563
	Appendix 14.2	564
	References	566
15	Local Dielectric Relaxation by Solvation Dynamics	571
	<i>R. Richert</i>	
15.1	Introduction	571
15.2	Experimental Techniques	572
15.3	Dielectric Aspects of Solvation	576
15.4	Solvation Dynamics	580
15.4.1	Dynamic Heterogeneity	583
15.4.2	Confined Systems	587
15.5	Summary	592
	References	594
16	Dielectric and Mechanical Spectroscopy – a Comparison	597
	<i>T. Pakula</i>	
16.1	Introduction	597
16.2	Principles of Mechanical Analysis	600
16.3	Linear Viscoelastic Behaviour	602
16.4	Experimental Procedures and Typical Data Analysis	606
16.5	Differences in Data Treatment	610
16.6	Typical Behaviour of Polymeric Materials	612
16.6.1	Melts of Linear Polymers	612
16.6.2	Block Copolymer Systems	614
16.6.3	Highly Branched Polymers	618
16.7	Suggested Analysis of Modulus Spectra	619
16.8	Concluding Remarks	620
	References	623

17	Dielectric Spectroscopy and Multidimensional NMR – a Comparison	625
	<i>R. Böhmer, F. Kremer</i>	
17.1	Introduction	625
17.2	Selected NMR Methods	626
17.2.1	Spin-Relaxation Techniques	627
17.2.2	Stimulated Two-Time Echo Spectroscopy	631
17.2.2.1	Translational Displacements	632
17.2.2.2	Reorientational Dynamics	636
17.2.3	Multidimensional Techniques	638
17.2.3.1	Frequency-Domain Spectroscopy	638
17.2.3.2	Multiple-Time Correlations	641
17.3	Combined Dielectric and NMR Studies: Recent Experimental Examples	646
17.3.1	Dynamics in ‘Simple’ Glass Formers: Glycerol vs Toluene	646
17.3.2	Motional Processes of a Triphenylene-Based Discotic Liquid Crystal	652
17.3.3	α - and β -Relaxations in Poly(methyl methacrylate)	657
17.3.4	Dynamic and Spatial Heterogeneity in Poly(vinyl acetate)	662
17.3.5	$2\text{Ca}(\text{NO}_3)_2 \cdot 3\text{MNO}_3$ as an Example for an Ion Conductor	668
17.3.6	Self-Diffusion in Polymeric Systems	671
17.4	Concluding Remarks	674
	Appendix 17.1	675
	Appendix 17.2	676
	References	679
 18	 Polymer Dynamics by Dielectric Spectroscopy and Neutron Scattering – a Comparison	 685
	<i>A. Arbe, J. Colmenero, D. Richter</i>	
18.1	Introduction	685
18.2	Quasielastic Neutron Scattering	686
18.2.1	Neutron Cross Sections and Correlation Functions	686
18.2.2	Neutron Instrumentation	688
18.2.2.1	Time of Flight Spectrometer	688
18.2.2.2	Backscattering Instrument	689
18.2.2.3	Spin Echo Spectrometer	690
18.2.3	Relation to Dielectric Spectroscopy	692
18.3.	Dynamic Structure Factors for the α - and β -Process	695
18.3.1	Local Processes	695
18.3.2	Diffusive (Q Dispersive) Processes	697
18.3.3	Merging of the α - and β -Processes	699
18.4.	Experimental Examples	699
18.4.1	Polybutadiene as a Showcase	699

18.4.2	Combined Dielectric and Neutron Investigations of the β -Process	703
18.4.3	α -Process on Polybutadiene	705
18.4.3.1	Temperature Dependence	705
18.4.3.2	Q-Dependence	706
18.4.3.3	Merging of α - and β -Processes	707
18.5.	Conclusions	714
	References	717
	For Further Reading	719
	Subject Index	721

2 Broadband Dielectric Measurement Techniques (10⁻⁶ Hz to 10¹² Hz)

F. Kremer · A. Schönhalz

2.1 Introduction

The complex dielectric¹ function $\varepsilon^*(\omega) = \varepsilon'(\omega) - i\varepsilon''(\omega)$ can be measured in the extraordinary broad frequency regime [1–13] from 10⁻⁶ Hz up to 10¹² Hz (in wavelength 3×10^{16} cm – 0.03 cm). To span this dynamic range different measurement systems based on different measurement principles have to be combined (Fig. 2.1). From 10⁻⁶ to 10⁷ Hz lumped circuit methods are used in which the sample is treated as a parallel or serial circuit of an ideal capacitor and an ohmic resistor. Effects of the spatial extent of the sample on the electric field distribution are neglected. With increasing frequency the geometrical dimensions of the sample capacitor become more and more important limiting this approach to about 10 MHz. In addition parasitic impedances caused by cables, connectors, etc. become important at frequencies >100 kHz.

Using distributed circuit methods (10⁷ Hz – 10¹¹ Hz) the complex dielectric function is deduced by measuring the complex propagation factor (in reflection or transmission). Both, waveguide as well as cavity techniques can be applied.

In the dynamic range between 10⁻⁶ Hz and 10¹⁰ Hz the complex dielectric function $\varepsilon^*(\omega)$ can also be deduced from a measurement of the time dependent dielectric function $\varepsilon(t)$. The latter is related to $\varepsilon^*(\omega)$ via a Fourier transformation

$$\varepsilon^*(\omega) - \varepsilon_\infty = \int_{-\infty}^{+\infty} \dot{\varepsilon}(t) e^{-i\omega t} dt \quad (2.1)$$

with $\varepsilon_\infty = \varepsilon'(\nu \approx 10^{11} \text{ Hz})$. For frequencies between 10¹⁰ Hz – 10¹² Hz (in wave numbers $\bar{\lambda} = \lambda^{-1} = 0.3 \text{ cm}^{-1} - 30 \text{ cm}^{-1}$, λ : wavelength in vacuum) quasioptical set-ups as polarizing Mach-Zehnder interferometers or oversized cavity resonators are employed. The latter has the advantage that it is suitable for very low loss materials and for measurements at low temperatures. The main experimental difficulty in this frequency range arises from the large coherence length of

¹ Dielectric data are also often presented in terms of the complex conductivity $\sigma^* = i\omega\varepsilon_0\varepsilon^*$ or the complex electrical modulus $M^* = 1/\varepsilon^*$. For dipolar relaxation processes ε^* is usually used while for systems with mainly conductive processes σ^* or M^* prevails.

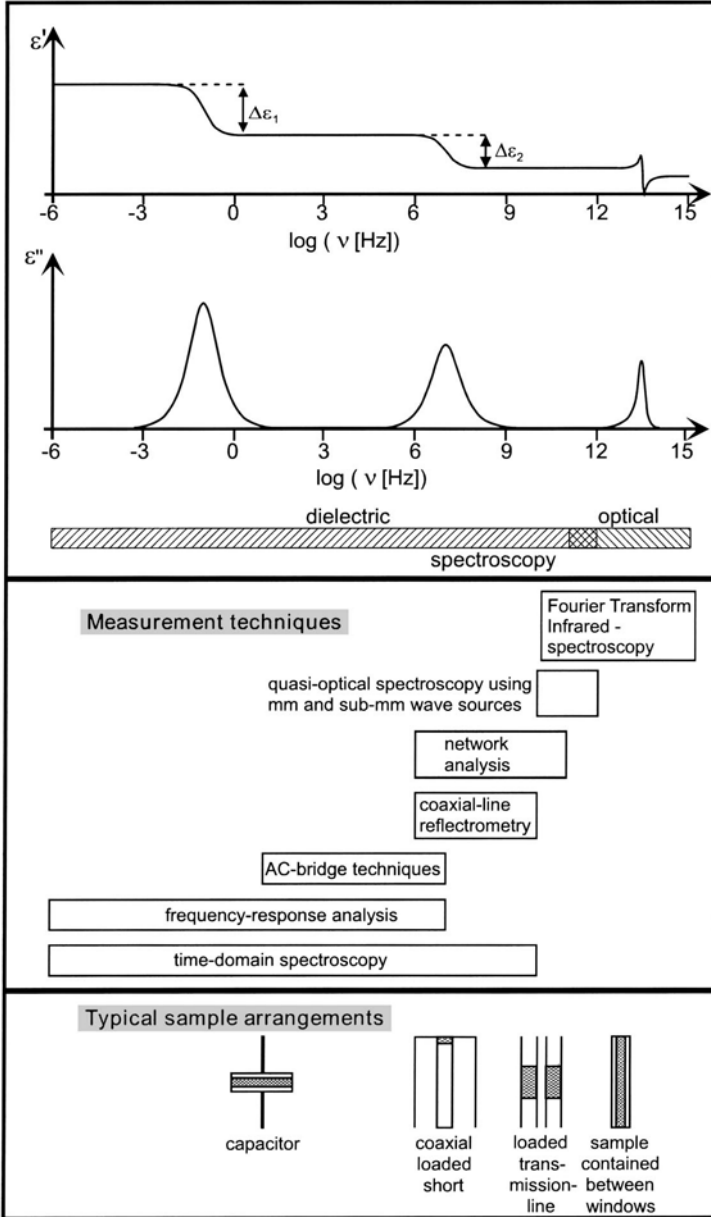


Fig. 2.1. Survey of measurement techniques used in the frequency range from 10^{-6} Hz – 10^{15} Hz

the mm- and sub-mm-wave sources (backward wave oscillators, klystrons, gun-oscillators, etc.) resulting in multiple standing wave patterns. Above 10^{11} Hz Fourier-Transform spectrometers are used. Due to the throughput and the multiplex advantages they are superior to arrangements using monochromators. More specialized dielectric techniques are discussed in Chaps. 14–16.

2.2

Measurements in the Frequency Domain from 10^{-6} Hz to 10^{11} Hz

For a capacitor C^* filled with a material under study the complex dielectric function is defined as

$$\varepsilon^*(\omega) = \varepsilon'(\omega) - i\varepsilon''(\omega) = \frac{C^*(\omega)}{C_0} \quad (2.2)$$

where C_0 is the vacuum capacitance of the arrangement. ω is the angular frequency with $\omega = 2\pi\nu = 2\pi T^{-1}$ with T as time for one period. $\varepsilon'(\omega)$ and $\varepsilon''(\omega)$ describe the real and imaginary part of the complex dielectric function. Using a sinusoidal electric field $E^*(\omega) = E_0 \exp(i\omega t)$ with the angular frequency ω and at field strengths within linear response (for most materials $E_0 \leq 10^6$ V cm $^{-1}$) the dielectric function can be derived by measuring the complex impedance $Z^*(\omega)$ of the sample

$$\varepsilon^*(\omega) = \frac{J^*(\omega)}{i\omega\varepsilon_0 E^*(\omega)} = \frac{1}{i\omega Z^*(\omega)C_0} \quad (2.3)$$

where $J^*(\omega)$ is the complex current density and ε_0 the permittivity of free space. To cover the frequency domain from 10^{-6} Hz up to 10^{11} Hz four different systems based on different measurement techniques are employed [14]: Fourier correlation analysis in combination with dielectric converters (10^{-6} Hz– 10^7 Hz) [15–17], impedance analysis (10^1 Hz– 10^7 Hz), RF-reflectometry (10^6 Hz– 10^9 Hz) [18, 19] and network analysis (10^7 Hz– 10^{11} Hz) [20, 21].

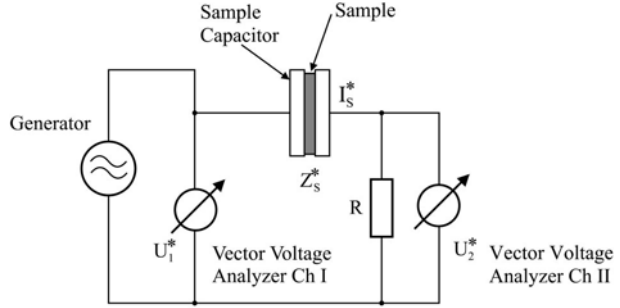
2.2.1

Fourier Correlation Analysis

The basic principle is shown in Fig. 2.2. Typically a sine wave voltage $U_1(t)$ with frequency $\omega/2\pi$ is applied to the sample by a generator, covering the frequency range from 10^{-6} Hz to 10^7 Hz 2 . The resistor R converts the sample current $I_s(t)$ into a voltage $U_2(t)$. $U_1(t)$ and $U_2(t)$ are analyzed with respect to the amplitudes and phases of their harmonic Fourier base waves $U_1^*(\omega)$ and $U_2^*(\omega)$ by two phase sensitive sine wave correlators. The complex sample impedance Z_s^* is calculated

² Fourier correlation analysis for frequencies $< 10^{-2}$ Hz becomes extremely time consuming. In this case measurements in the time domain are advantageous.

Fig. 2.2. Scheme of a Fourier correlation analyzer. Taken from [10] with permission



from the measured data by

$$Z_s^*(\omega) = \frac{U_s^*(\omega)}{I_s^*(\omega)} = R \left(\frac{U_1^*(\omega)}{U_2^*(\omega)} - 1 \right) \quad (2.4)$$

where $U_s^*(\omega)$ and $I_s^*(\omega)$ are the sample voltage and current. The response of the correlator to the signal $U_i(t)$ are the two components

$$U_j'(\omega) = \frac{1}{NT} \int_0^{NT} U_j(t) \sin(\omega t) dt \quad (2.5a)$$

and

$$U_j''(\omega) = \frac{1}{NT} \int_0^{NT} U_j(t) \cos(\omega t) dt \quad (2.5b)$$

with $j = 1, 2$. U_j' is the in phase component and U_j'' the orthogonal (90° shifted or quadrature) component of the harmonic base wave. N is the number of periods with duration $T = 2\pi/\omega$ measured by the correlator³.

For technical implementation of Fourier correlation analysis one has to realize Eq. (2.5a/b). This is done in conventional systems by analog components. From the main oscillator, the reference sin and cos signals are created by a 90°

³ For readers having no experience in Fourier analysis, the following may be helpful in order to understand the correlator operation. If $U_i(t)$ is a sine wave with amplitude U_0 and phase angle ϕ

$$U_i(t) = U_0 \sin(\omega t + \phi) \quad (2.5c)$$

for the product holds

$$U_j(t) \sin(\omega t) = U_0 \sin(\omega t + \phi) \sin(\omega t) = \frac{U_0}{2} (\cos(\phi) - \cos(2\omega t + \phi)) \quad (2.5d)$$

As the integration in Eqs. (2.5a/b) is applied over a number of periods T , the second term in Eq. (2.5d) averages to zero. Therefore, the integral (Eq. 2.5a) becomes $U_i'(\omega) = U_0 \cos(\phi)/2$. With the same arguments, $U_i''(\omega) = U_0 \sin(\phi)/2$. If $U_i(t)$ is a signal of arbitrary type like, e.g., a sine wave superimposed with noise, it can be transformed into its Fourier representation consisting of a number of sine waves. For each frequency component both terms of

phase shifter, multiplied with the test signal and finally integrated. Modern systems digitize the test signal and approximate numerically Eqs. (2.5a/b) by a signal processor avoiding by that the drift and the non-linearity of analogue components. Fourier correlation analyzers are commercially available as frequency response analyzers or lock-in amplifiers from Agilent Technologies, EG & G, Novocontrol, Solartron and Stanford Research Systems.

2.2.2

Dielectric Converters in Combination with Fourier Correlation Analysis

For measurements, the setup from Fig. 2.2 with the resistor R suffers from several limitations and therefore a dielectric converter is usually used for conversion of the sample current to $U_2(t)$ [15–17]. The Fourier correlation analysis is done either by an additional frequency response analyzer or lock-in amplifier. The dielectric converter uses a broadband electrometer amplifier with variable gain (Fig. 2.3) in order to match the huge impedance range required for dielectric measurements which typically covers about 13 orders of magnitude over the frequency range from 1 mHz to 10 MHz.

If $Z_X^*(\omega)$ is a variable impedance which can be changed in resistance and capacitance, the sample impedance $Z_S^*(\omega)$ of a direct measurement is given by

$$Z_S^*(\omega) = \frac{U_{1S}^*(\omega)}{I_S^*(\omega)} = -\frac{U_{1S}^*(\omega)}{U_{2S}^*(\omega)} Z_X^*(\omega) \quad (2.6)$$

The accuracy of the measurement according to Eq. (2.6) is limited due to amplitude and phase errors in the current to voltage converter and the correlators.

Further improvement results from a comparison of the sample capacitor with a variable converter internal reference capacitor with impedance $Z_R^*(\omega)$. If a reference measurement under the same conditions (especially for the current-to-voltage converter) is made

$$Z_R^*(\omega) = \frac{U_{1R}^*(\omega)}{I_R^*(\omega)} = -\frac{U_{1R}^*(\omega)}{U_{2R}^*(\omega)} Z_X^*(\omega) \quad (2.7)$$

(continued Footnote 3)

Eq. (2.5d) are frequency dependent and will average out to zero if the integration is performed over many periods. The only frequency component which does not average out is the one at the correlator frequency $\omega/2\pi$. Therefore, the correlator phase-sensitively detects the base Fourier component contained in any input signal. All other contributions are averaged out. The quality of suppression increases with averaging time determined by the number N of integration periods. Therefore, this technique generally is used in precision instrumentation like lock-in amplifiers for noise suppression or if small signals have to be measured, e.g., measuring a signal in the nV range superimposed by a 1000 times higher noise signal is possible. These properties are the basis for the many advantages sine wave correlation systems offer with respect to non-correlated analysis systems like, e.g., time domain techniques or bridges where any noise or non-linear response usually strongly contributes to the measured signal. As a further advantage of sine wave correlation systems, higher harmonic signal components may readily evaluated by increasing the number N in Eqs. (2.5a/b).

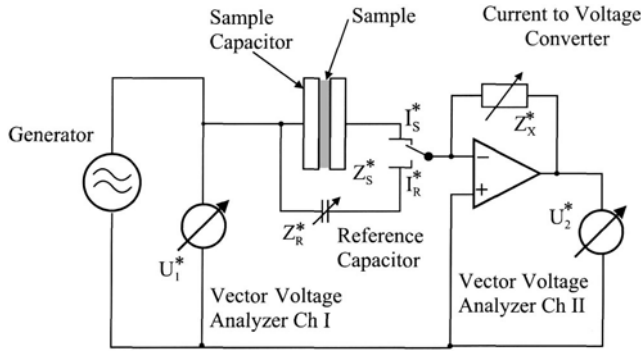


Fig. 2.3. Scheme of a dielectric converter in the low-frequency range with electrometer amplifier and variable reference capacitor

Combining Eqs. (2.6) and (2.7) yields

$$Z_S^*(\omega) = \frac{U_{1S}^*(\omega)}{U_{2S}^*(\omega)} \cdot \frac{U_{2R}^*(\omega)}{U_{1R}^*(\omega)} \cdot Z_R^*(\omega) \quad (2.8)$$

which is independent of linear amplitude or phase errors of the correlators, as they enter both in the sample and reference measurement and therefore are canceled out. This works in practice only if the sample impedance and reference impedance are nearly in the same range. In this case the accuracy is limited only by the correlator resolution. Therefore, the reference capacity for some dielectric converters can be adjusted, e.g., in 64 steps between 25 pF and 2 nF [16, 17].

For frequencies < 100 kHz (Fig. 2.3) a low current operational amplifier can be used. Such amplifiers are available for extreme low input currents down to some fA. On the other hand, these amplifiers are limited to low frequencies. Therefore, some dielectric converters [16] use for frequencies between 10^5 Hz and 10^7 Hz a different technique of current to voltage conversion (Fig. 2.4). The a.c. voltage of the generator drives a buffer amplifier which decouples the sample. In contrast to Fig. 2.3 the active current-to-voltage converter is replaced by a passive variable impedance Z_X^* . For ideal components and neglecting inductive effects one finds

$$Z_S^*(\omega) = Z_X^* \left(\frac{U_{1S}^*(\omega)}{U_{2S}^*(\omega)} - 1 \right) \quad (2.9)$$

With this equation the reference technique can be applied in a similar manner as above.

Dielectric converters enable one to measure highly accurate results for capacitive samples with not too high losses and only low d.c. conductivity. For high d.c. conductivity or high dielectric losses, the sample impedance is mainly resistive and often not in the range of the reference capacitors. For such kinds of samples, impedance analysis as discussed below often yields better results.

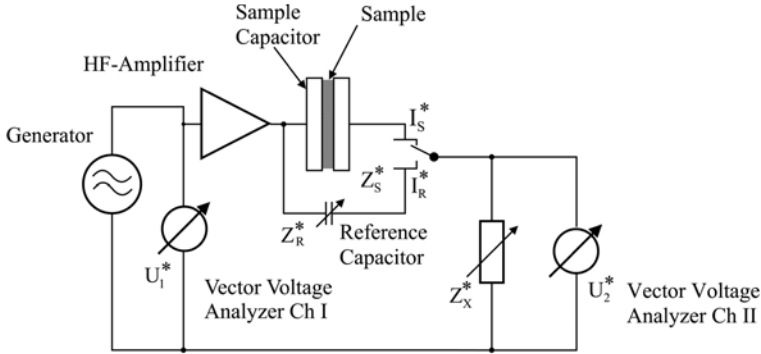


Fig. 2.4. Scheme of a dielectric converter in the high-frequency range with variable impedance and variable reference capacitors

Using state of the art equipment, an accuracy in $\tan \delta = \frac{\epsilon''}{\epsilon'}$ better than 10^{-4} can be achieved over a broad frequency range. The limits for high impedance and low capacitance are around $10^{14} \Omega$ and 1 pF. Low impedances down to about 100Ω may be measured too, but only with reduced accuracy due to the limitations of the reference technique.

Dielectric converters in combination with frequency response analyzers (FRAs) or lock-in amplifiers are commercially available from, e.g. Novocontrol and Solartron.

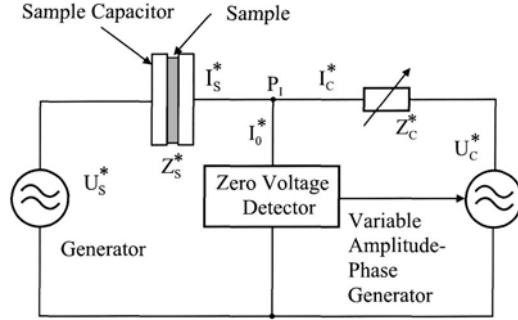
Similar systems with electrometer amplifiers but without reference technique are commercially available from, e.g. Micromet, Rheometrics, Seiko, and TA Instruments.

2.2.3

Impedance Analysis

Two kind of techniques have to be distinguished. The first is the I-V method referring to the direct phase sensitive measurement of the sample current and voltage. It is similar to the dielectric converter techniques described above. Instead of an electrometer amplifier, a broadband current to voltage converter is used without reference technique.

An alternative technique is the a.c. impedance bridge shown in Fig. 2.5 which consists of the sample capacitance $Z_S^*(\omega)$ and the adjustable compensation impedance $Z_C^*(\omega)$. On the left hand side of the bridge, the generator drives the sample with the fixed and known a.c. voltage $U_S^*(\omega)$ which causes the current $I_S^*(\omega)$ to flow into P_L . On the right hand side of the bridge, the variable amplitude-phase generator (VAPG) feeds the current $I_C^*(\omega)$ through the compensation impedance $Z_C^*(\omega)$ into P_L . The bridge will be balanced, if $I_S^*(\omega)$ equals $-I_C^*(\omega)$ which corresponds to $I_0 = 0$. Any deviation is detected by the zero voltage detector which changes the amplitude and phase of the variable amplitude phase generator

Fig. 2.5. Scheme of an impedance bridge

(VAPG) as long as $I_0 \neq 0$. In the balanced state, the sample impedance is calculated as

$$Z_s^*(\omega) = \frac{U_s^*(\omega)}{I_s^*(\omega)} = -\frac{U_s^*(\omega)}{U_c^*(\omega)} Z_c^*(\omega) \quad (2.10)$$

Impedance bridges or analyzers which are not specially designed for dielectric measurements typically work from 10 Hz to 10 MHz. For dielectric samples, they suffer from the inverse problem of the dielectric converter systems as optimized for low (typically 10 m Ω) to medium impedance with a limit of about 100 M Ω . This limit is reached for typical dielectric samples (100 pF, $\tan \delta = 0.01$) at about 1 kHz. The typical accuracy in $\tan \delta$ is $< 10^{-3}$. Here, a.c.-impedance analyzers or bridges offer a suitable and inexpensive solution for applications which do not require a large frequency and a high impedance range. Due to the fact that the measurement time is lower compared with systems using reference techniques, a.c.-impedance bridges are particularly suited for measurements on materials with time-dependent dielectric properties (monitoring of chemical reactions, characterization of phase transitions, etc.). Impedance analyzers or bridges are commercially available from several manufactures like, e.g., Agilent Technologies, Novocontrol, Solartron, QuadTech, and Wayne Kerr.

2.2.4

Integrated Dielectric Analyzers

Systems combining the advantages of Fourier correlation analysis, with dielectric converters and impedance analysis, are the latest progress in dielectric instrumentation [17]. These incorporate in one device a direct digital synthesizer generator, two digital precision sine wave correlators, a state of the art dielectric converter, and a direct I–V current to voltage converter for the low impedance range. For typical dielectric capacitive samples, the dielectric converter is used as a precision front end. In this mode broadband phase and $\tan \delta$ accuracy $< 3 \times 10^{-5}$ is available. If the sample impedance is out of the dielectric converter reference capacitor range, the front end is switched automatically to the I–V converter.

With this concept an overall impedance range from 0.01Ω to $10^{14} \Omega$ is covered with one single device. The wide impedance range results in a wide capacity range as well. Ultralow capacities can be measured with high precision to a limit of 1 fF (10^{-15} F). High resolution dielectric analyzers are commercially available from Novocontrol.

2.2.5

Samples and Sample Cells for Frequencies <10 MHz

Typical samples have a thickness of $\approx 50 \mu\text{m}$ and diameter of 10 mm resulting in a capacitance of 69.5 pF for a value of $\epsilon' = 5$. It is essential to measure the sample temperature in the immediate sample neighborhood and to keep the sample temperature as constant as possible. As sample geometry a parallel plate capacitor arrangement is commonly used.

For the techniques discussed so far, a sample cell is required which is connected by BNC cables to the impedance plugs of the measurement system. Due to the parasitic inductance of the lines and connectors the high frequency limit is reached at about 1 MHz. At the lower frequency range end, very small currents down to fA have to be measured. In this region, the measurement may be obscured by noise pick up and piezoelectric charges in the cables due to mechanical stress. Therefore a sophisticated cell and cable design optimized for both high frequency performance and best insulation over the entire frequency and temperature range is required.

As an alternative, active sample cells may be used where the electrometer amplifier and the reference capacitors are integrated in the sample cell head avoiding BNC cables in the sample impedance circuit [16, 17]. For such systems, the active cell can be considered as an analyzer system and the accuracy and range specifications apply in parallel. This is in contrast to systems with BNC cables and passive sample cells which are usually specified at the analyzer BNC input plugs neglecting cable and cell effects.

2.2.6

RF Reflectometry

Coaxial line reflectometry [14, 18, 19] can be employed at frequencies from 1 MHz to 10 GHz. In contrast to the low frequency techniques already described, above 1 MHz the measurement cables significantly contribute to the sample impedance. Above approximately 30 MHz standing waves arise in the line and a direct measurement of the sample impedance completely fails. This can be avoided by application of microwave techniques taking the measurement line as the main part of the measured impedance into account. Therefore, precision lines and sample cells with defined propagation constants are required. A schematic of the measurement technique is shown in Fig. 2.6. The sample capacitor is used as the termination of a precision coaxial line.

The complex reflection factor $r^*(l)$ of the line (length l) depending on the sample impedance is measured with a microwave reflectometer. For this pur-

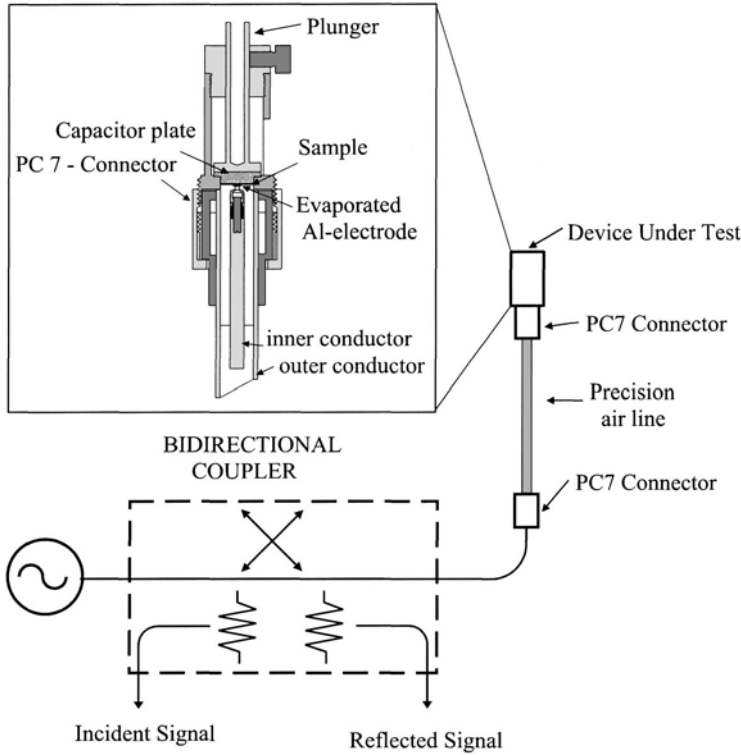


Fig. 2.6. Scheme of the coaxial line reflectometer with sample head

pose, the incoming and reflected waves are separated with two directional couplers and are measured in amplitude and phase. $r^*(x)$ is defined as the ratio of the voltages (or electrical fields) of the reflected and the incoming wave of the line. It depends on the location of the measurement along the line

$$r^*(x) = \frac{U_{\text{refl}}^*(x)}{U_{\text{inc}}^*(x)} \quad (2.11)$$

For an ideal line, $r^*(l)$ which is measured by the reflectometer is related to the reflection factor $r^*(0)$ at the beginning ($l = 0$) of the line by

$$r^*(l) = r^*(0) \exp[2l(\alpha + i\beta)] \quad (2.12)$$

where α is the attenuation coefficient ($\alpha = \frac{2\pi n''}{\lambda}$) and β the phase (or propagation) coefficient ($\beta = \frac{2\pi n'}{\lambda}$). From Eq. (2.11) the sample impedance is calculated as

$$Z_S^*(\omega) = Z_0 \frac{1 + r^*(l)}{1 - r^*(l)} \quad (2.13)$$

where Z_0 is the wave resistance of the line. As can be seen from Eq. (2.13), the measurement range is limited to sample impedances Z_s^* being in the region of Z_0 .

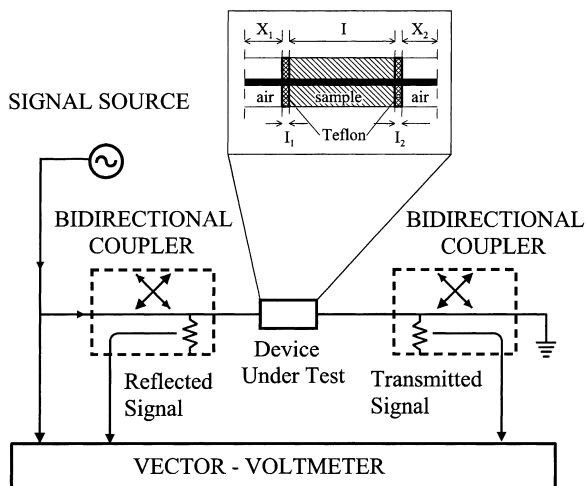
In practice, the lines are not ideal and sophisticated calibration procedures have to be applied and low loss precision lines matching the output resistance of the reflectometer are required. The line parameters α, β must be homogeneous over the whole line and also independent of temperature, as the calibration generally can only be carried out at room temperature. The same criteria mentioned for the line apply to the sample cell. Therefore, an additional calibration which eliminates the influence of internal impedances in the sample cell is necessary. In order to keep the waveguide as short as possible, the cryostat with the sample has to be closely mounted at the frontend of the analyzer. Coaxial line reflectometers cover the frequency range between 1 MHz and ~ 3 GHz. Based on Agilent RF-impedance analyzers an accuracy in $\tan \delta < 10^{-2}$ can be realized (HP 4291 B: 1 MHz–1.8 GHz, HP 4991 A: 1 MHz–3 GHz). Sample cells, precision extension line, and dedicated cryostats are commercially available.

2.2.7

Network Analysis

At frequencies above 1 GHz network analysis [20, 21] can be used (Fig. 2.7) in which not only the reflected wave but also the wave transmitted through a sample is analyzed in terms of phase and amplitude. This allows the frequency range to be extended up to 100 GHz. However, with increasing frequency and hence decreasing dimensions of coaxial lines or waveguides, the calibration procedure becomes cumbersome. With network analyzers based on the Agilent 8510 an accuracy in $\tan \delta < 10^{-2}$ is possible.

Fig. 2.7. Scheme of the vectorial network analyzer with transmission cell



2.2.8

Impedance and Frequency Ranges Overview

It is convenient to represent the sample impedance as a capacitor C_s related to ϵ' in parallel circuit with a resistor R_s related to ϵ'' and the material d.c. conductivity σ_{dc} .

The serial impedance of the sample capacitor is

$$Z_s^* = \frac{U_s}{I_s' + i I_s''} = \frac{1}{i C_0 \left(\omega \epsilon^* - i \frac{\sigma_{dc}}{\epsilon_0} \right)} \quad (2.14)$$

The parallel impedance Z_p^* is defined with respect to each sample current component I' and I'' as

$$Z_p^* = \frac{U_s}{I_s'} + i \frac{U_s}{I_s''} \quad (2.15)$$

Combination with Eq. (2.14) yields

$$Z_p' = \frac{1}{\left(\omega \epsilon'' + \frac{\sigma_{dc}}{\epsilon_0} \right) C_0} \quad \text{and} \quad Z_p'' = \frac{1}{\omega \epsilon' C_0} \quad (2.16)$$

The dark shaded area in Fig. 2.8 shows Z_p' and Z_p'' for a typical dielectric sample with capacity C_s around 100 pF, loss factor $\tan \delta$ in the range from 10⁻⁴ to 1 and $\sigma_{dc} = 0$. Both Z_p' and Z_p'' show the typical $1/\omega$ increase with decreasing frequency for capacitive samples. Due to the broad frequency regime, the impedance range becomes extraordinary large. For high losses with e.g. $\tan \delta = 1$, the real and imaginary parts of Z_p^* become equal. With decreasing ϵ'' and $\tan \delta$, Z_p' increases linearly shifting for constant $\tan \delta$ and C_s to higher impedances.

For comparison, the typical impedance ranges of the several analysis techniques are shown, too. The outer border lines of each range correspond to 10% accuracy for each Z_p component which may be seen as a reasonable limit for the usable range. For impedance points within this range, accuracy linearly increases from 10% with the logarithmic distance to the border line until the systems base accuracy is reached. For example, if the impedance point is located within the range two orders of magnitude apart from a border line, the accuracy would be 0.1%. The total impedance limit (100% error) would be reached for impedance points one order of magnitude outside the marked areas.

As can be seen from Fig. 2.8, the capacitive component Z_p'' related to ϵ' is generally lower in impedance (for $\tan \delta < 1$) and therefore it can be measured with all techniques within their frequency range. At the high frequency end, the RF reflection and network analyzers suffer from too low sample impedances. Therefore, one usually reduces the sample capacity by increasing the sample thickness or decreasing its diameter. Typical values are within 1 pF–10 pF for

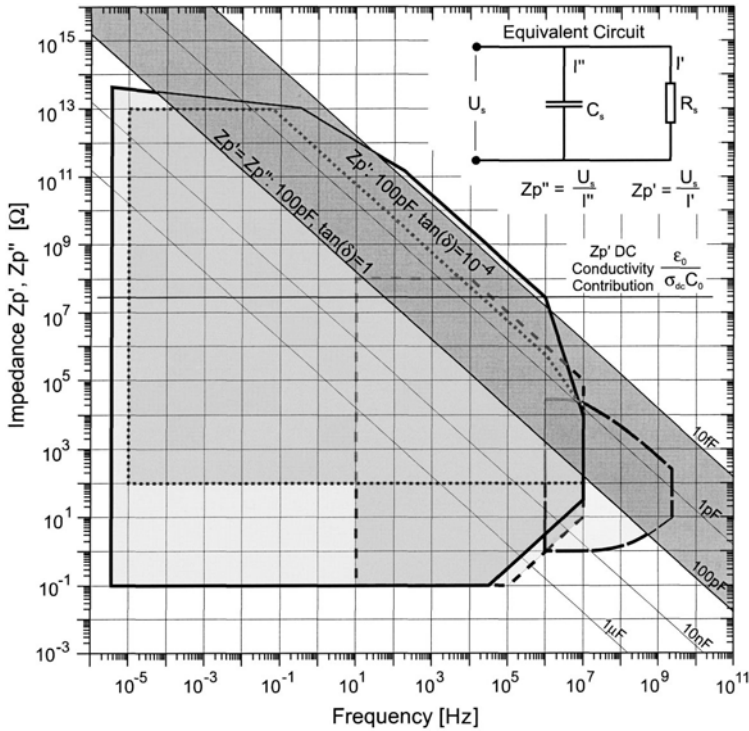


Fig. 2.8. Ranges (10% accuracy) of complex impedance [Z_p' , Z_p''] and frequency for a dielectric sample (100 pF, $\tan \delta$ from 10^{-4} to 1) represented by a parallel circuit of a capacitor and resistor (inlet). Different measurement systems are compared: --- Impedance analyzer, Dielectric converter with Fourier correlation analyzer, — Dielectric analyzer, —••— RF reflectometer and network analyzer

optimized performance at the high frequency end. This is only possible at the further cost of decreased accuracy at the RF low frequency range end (1 MHz).

For the Z_p' component related to ϵ'' , the situation becomes more critical with decreasing losses. For $\tan \delta = 10^{-2}$, the 10% borderline applies for all systems at frequencies >10 MHz, meaning that low loss dielectrics cannot be measured at high frequencies with the broadband techniques presently available. The general impedance analyzers reach their Z_p' 10% limit below about 1 kHz. With further decreasing losses $<10^{-3}$, the dielectric converters reach their 10% border line as well. Lower loss samples down to 10^{-4} can be measured with the dielectric analyzer from about 10 Hz–100 kHz.

The d.c. conductivity contributes, according to Eq. (2.16), a frequency independent part $\frac{\epsilon_0}{\sigma_{dc} C_0}$ to Z_p' which is drawn as a horizontal line in Fig. 2.8. This line confines the dielectric contribution $\frac{1}{\omega \epsilon'' C_0}$ to Z_p' at high impedance. As Z_p'' increases to low frequencies with $1/\omega$, there is always a turn over frequency

$\omega_t = \frac{\sigma_{dc}}{\epsilon_0 |\epsilon^*|}$ where the sample impedance changes from capacitive to resistive

characteristics. Depending on the material, σ_{dc} and with this ω_t may vary over many orders of magnitude especially if measured at different temperatures. For high d.c. conductivity, Z_p' may become small at low frequencies. This causes problems for systems based on dielectric converters, as their reference capacitor technique usually fails in this range. It is then more convenient to use the general impedance techniques or the dielectric analyzer which offer high accuracy over the entire impedance range both for low loss and high conductive samples.

2.3

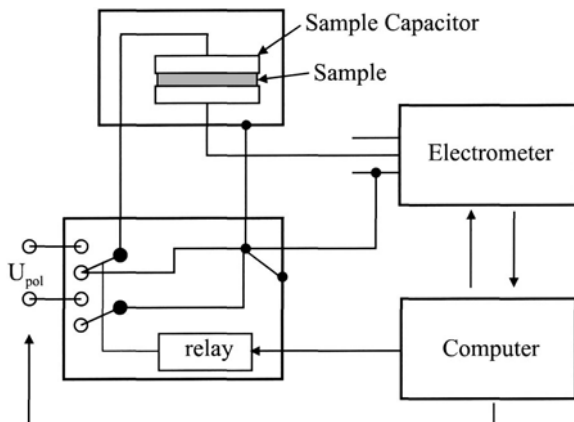
Measurement Systems in the Time Domain from 10⁻⁶ Hz to 10¹⁰ Hz

In the dynamic range from 10⁻⁶ Hz to 10¹⁰ Hz the complex dielectric function can also be determined by measuring the time dependence of the (d.c.)-polarization current of a (loaded) sample capacitor [22–26]. The experimental approach is simple and because of the multiplex advantage less time consuming than measurements in the frequency domain. However, this is possible only on the expense of a reduced accuracy. In analogy to Eq. (2.1)

$$\epsilon(t) = \frac{C(t)}{C_0} \quad \text{and} \quad \frac{d\epsilon(t)}{dt} = \frac{I(t)}{C_0 U_{Pol}} \quad (2.17)$$

hold where $I(t)$ is the polarization current and U_{Pol} the polarizing voltage. C_0 is the capacitance of the empty capacitor. According to Eq. (2.13) the time dependent dielectric function is determined from a measurement of the time dependence of the (d.c.)-polarization current $I(t)$. (Depolarization experiments are often more convenient because they are not disturbed by a d.c.-conductivity contribution. The driving external electric field is zero.) A typical experimental

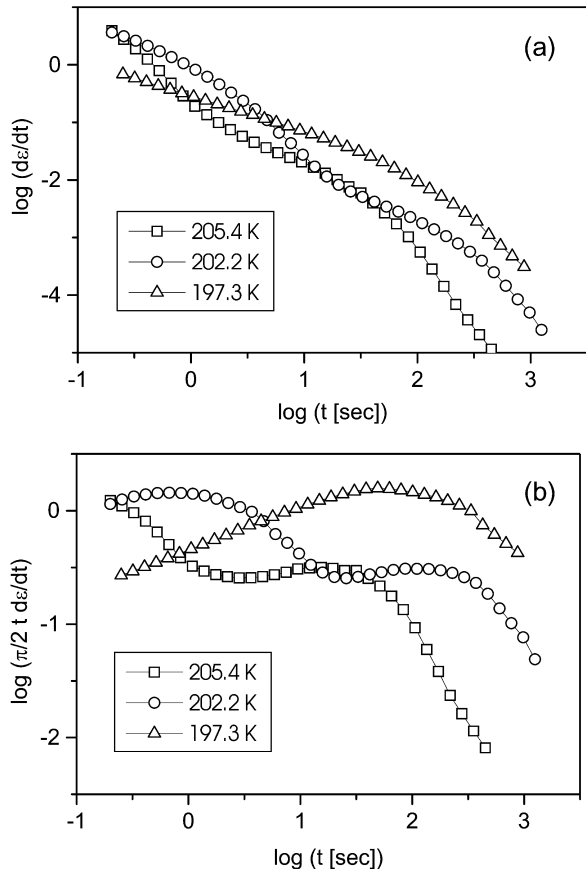
Fig. 2.9. Scheme of the time-domain set-up for frequencies up to 100 Hz with mechanical switches and up to 10⁴ Hz with electronic switches [22]



set-up for a low frequency (≤ 10 kHz) time domain spectrometer is shown in Fig. 2.9. In order to achieve a high resolution in $\tan \delta \leq 10^{-3}$ three requirements must be fulfilled: (i) a sensitive electrometer covering a wide dynamic range from 10^{-3} A to 10^{-15} A (Keithley 617 or Keithley 6517), (ii) low noise triaxial cables with an insulation resistance of $\geq 10^{15} \Omega$, and (iii) earth loops caused by different ground sites must be avoided.

An example of a time domain measurement is presented in Fig. 2.10a/b for the glass transition range of poly(propylene glycol) [27]. To obtain the complex dielectric function in the frequency domain a Fourier transform would be necessary. As a rough estimation of $\epsilon^*(\omega)$ the function $\pi/2t \, d\epsilon/dt$ can be used, which is called the Hamon-transformation [23]. This representation of the time domain measurements shows maxima for relaxation processes similar as for the loss part of the complex dielectric function ϵ'' . An example is given for poly(propylene glycol) in Fig. 2.10b. With the relation $\omega = 0.2\pi \, t^{-1}$ the Hamon-transformation can be directly compared with ϵ'' with respect to the maximum position and high frequency behavior of a relaxation process. At the low frequency side of a relaxation process these two functions behave differently. An

Fig. 2.10.
a Depolarization current vs time at different temperatures as indicated. b Data after a Hamon transformation. Sample: poly(propylene glycol)



exact transformation of time domain data into the frequency domain requires a Fourier transform which can be done numerically (see for instance [28, 29]). Some approximative methods are discussed in [30, 31].

An alternative approach to the dielectric polarization as a function of time is based on the recent development of electrical modulus $M(t)$ measurements. Under the condition of a constant charge (instead of the usual constant field case), the voltage across the capacitor is proportional to the polarization and to the modulus $M(t)$, with $M^*(\omega) = 1/\epsilon^*(\omega)$. The experimental setup creates a step function in the dielectric displacement $D(t)$, while the voltage $U(t) \sim M(t)$ across the capacitor is measured [24a]. In order to assure a constant D for $t \geq 0$ the setup should not allow any external current flow within the circuit, i.e., a high input resistance of the voltmeter is essential. The step function in $D(t)$ at $t = 0$ is realized by connecting a voltage source with $U = U_0$ to the discharged capacitor for a short period of time. Figure 2.11 displays a schematic diagram of such a setup which employs the guarding technique in order to minimize cable capacitance effects. With a Keithley 6517 electrometer serving as voltmeter with guarded input ($R_{in} \geq 200 \text{ T}\Omega$, $C_{in} \approx 2 \text{ pF}$, $I_{bias} \leq 2 \text{ fA}$) and as voltage source, the polarization can be measured for times 10^{-3} s to 10^6 s .

The modulus technique is advantageous in several ways. For dielectric relaxation measurements at long times one can exploit the fact that $M(t)$ decays faster than $\epsilon(t)$ under identical conditions. The time scale relation for KWW type responses is given by

$$\tau_M = \left(\frac{\epsilon_\infty}{\epsilon_s} \right)^{\beta_\epsilon} \tau_\epsilon = \left(\frac{M_s}{M_\infty} \right)^{\beta_M} \tau_\epsilon, \quad \beta_M \approx 0.8 \beta_\epsilon \quad (2.18)$$

where $M_s = \epsilon_s^{-1}$ and $M_\infty = \epsilon_\infty^{-1}$ denote the electric modulus in the limits $t \rightarrow \infty$ and $t \rightarrow 0$, respectively [24b]. For the example of a polymer system shown in Fig. 2.12, this feature allowed access of an average time scale for dipole fluctuations $\log(\tau[\text{s}]) = 7.53$ (1.1 years). For ionic conductors, even severe cases of electrode polarization give rise to only small effects in the $M(t)$ signal [24c]. Furthermore, the technique has allowed dielectric hole-burning measurements to be applied to vitreous ionic conductors (see Chap. 14 on non-resonant dielectric hole burning spectroscopy).

Dielectric measurements in the time domain can be extended to high frequencies covering the range from 10^5 Hz to $2 \times 10^{10} \text{ Hz}$ [26]. As in the reflec-

Fig. 2.11. Schematic diagram for measuring the dielectric modulus $M(t)$ in the time domain

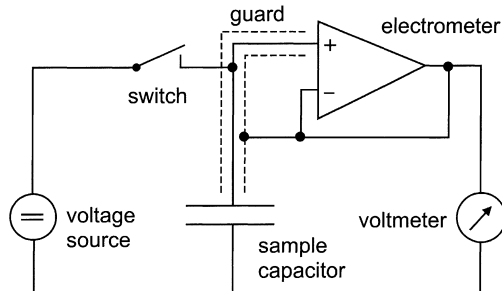
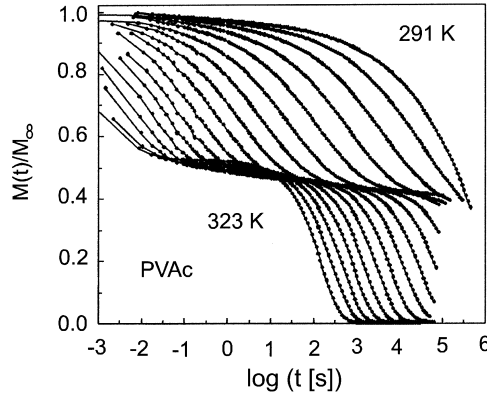


Fig. 2.12. Experimental results for the dielectric modulus $M(t)$ for poly(vinyl acetate) at temperatures from 291 K to 323 K in steps of 2 K. The contributions above and below the plateau at ≈ 0.5 reflect the α -process and d.c.-conductivity, respectively



tometer approach (see below) the sample is located at the end of a coaxial transmission line. A rectangular step voltage is sent to the sample cell and the reflected pulse is measured by use of a fast digital oscilloscope. Hence the accessible frequency range is determined by the width and rise time of the incident pulse. The resolution of these spectrometers is smaller than that of the equipment which is based on network analyzers in the frequency domain.

2.4

Quasioptical Set-Ups in the Frequency Range from 10^{10} Hz to 10^{12} Hz

In the frequency regime between 10^{10} Hz and 10^{12} Hz (mm- and sub-mm-waves) on the low frequency side ($< \sim 100$ GHz) coaxial and waveguide techniques are used and above 300 GHz ($\tilde{\lambda} = 10 \text{ cm}^{-1}$) Fourier-Transform spectroscopy is employed. The intermediate range is bridged by “quasioptical”-methods [32–35] using unguided monochromatic electromagnetic waves propagating through “free space”. As radiation sources typically backward-waveoscillators (BWO) are applied because of their tunability in frequency (in contrast to gunoscillators or clystrons which are narrow banded). The scheme of a quasioptical mm- and sub-mm-wave spectrometer based on a polarizing double beam Mach-Zehnder interferometer is shown in Fig. 2.13. The radiation of the BWO is emitted by a horn-antenna and focused by Teflon lenses. For measurements of the transmission $T(\omega)$ one arm of the interferometer is blocked. A set of two lenses is used to focus the radiation at the position of the sample in the cryostat or oven. The radiation is detected by a Golay cell or a (pumped) Helium InSb-bolometer. To reduce the noise level a chopper in the beam path and phase sensitive detection is applied. Two consecutive frequency sweeps with and without the sample at each temperature are made leading to a precise determination of the transmission coefficient T . Values as low as $T = 10^{-7}$ can be detected.

To record the phase shift $\varphi(\omega)$ the instrument works in the interferometer mode. For that the beam is split using a 45° wire-grid. In the second arm of the spectrometer a similar set of lenses is mounted in order to make the optical path

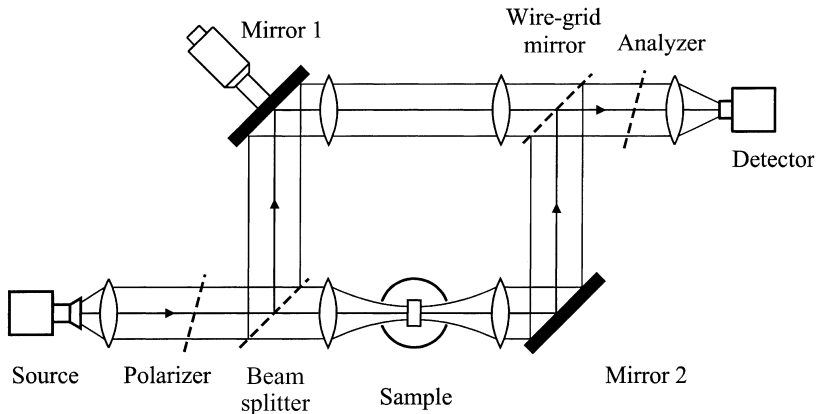


Fig. 2.13. Scheme of the polarizing quasi-optical Mach-Zehnder interferometer for the mm- and sub-mm-wave regime

length identical to that of the first arm (without sample). After combination of the two beams by another 45° -wire-grid the radiation is detected. Using the movable mirror in the reference arm the phase shift $\varphi(\omega)$ of the sample is measured. For an isotropic sample layer the complex transmission coefficient $t^* = \sqrt{T(v)} \exp(i(v))$ is determined by the following well known relationships [32, 33]

$$\begin{aligned}
 T &= E \frac{(1-R)^2 + 4R \sin^2 \psi}{(1-RE)^2 + 4RE \sin^2(N+\psi)} \\
 \varphi &= N + \arctan \frac{ER \sin 2(N+\psi)}{1-ER \cos 2(N+\psi)} + \arctan \frac{b}{a^2+b^2+a} - \arctan \frac{b}{a+1} \\
 E &= \exp\left(-\frac{4\pi n''d}{\lambda}\right), \quad N = \frac{4\pi n'd}{\lambda} \\
 R &= \frac{(a-1)^2 + b^2}{(a+1)^2 + b^2}, \quad \psi = \arctan \frac{2b}{a^2+b^2-1} \\
 n' + in'' &= \sqrt{\epsilon^* \mu^*}, \quad a + ib = \sqrt{\frac{\mu^*}{\epsilon^*}}
 \end{aligned} \tag{2.19}$$

where λ is the wavelength of the radiation, d the sample thickness, and ϵ^* and μ^* the complex dielectric permittivity and magnetic permeability of the substance, respectively.

A principal difficulty for measurements in the mm- and sub-mm-wave frequency range using monochromatic sources results from the high coherence length of the radiation. This gives rise to multiple interference patterns which may spoil the experimental accuracy. Phase-shakers can be used to circumvent

this problem [32]. For the above described Mach-Zehnder set-up the minimum detectable value for ϵ'' is limited by the maximum thickness of the sample only. A typical minimal value for the measurement of $\tan \delta$ is 10^{-2} .

An alternative experimental approach in the mm- and sub-mm-wave regime are oversized cavities [36, 37] with typical dimensions being much larger than the wavelength: using high reflectivity walls and “mode stirrers” an isotropic and homogeneous field is generated on a time-averaged basis. The usefulness of the device arises from several advantages. For absorption measurements of gaseous samples the oversized cavity serves as a long-path absorption cell, thus providing a manifold increase in sensitivity compared to a single-path set-up. The photon-storing capability of the device can also be used for measuring the resistivity of metals. In this case the walls of the oversized cavity are partly composed of the materials under study. For inhomogeneous samples, the main advantage arises from the fact that such measurements are only sensitive to the absorption loss in the specimen and not to scattering and reflection losses. Thus, specimen preparation in a plane-parallel form is not necessary, as it is for quasioptical methods. It therefore allows for the quantitative study of a substantial body of materials, including powders, thin films, and composites. This applies to the whole mm- and sub-mm-wave spectrum. For low temperature absorption

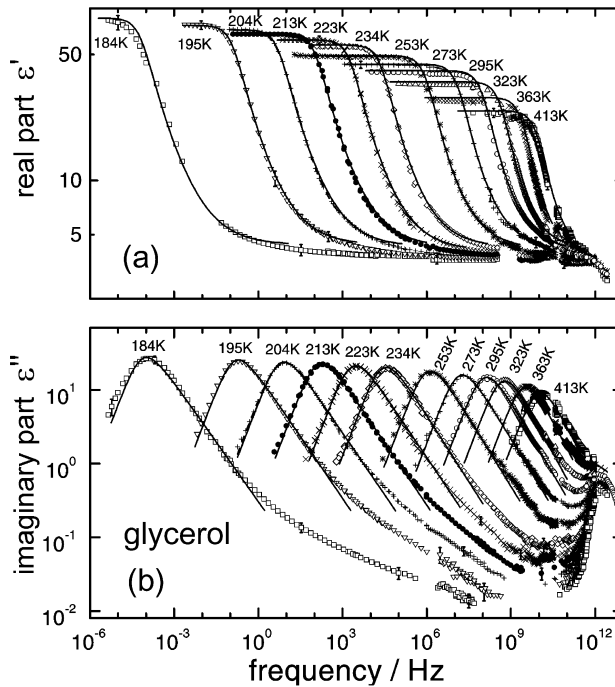


Fig. 2.14. Dielectric spectra: **a** real part ϵ' ; **b** imaginary part ϵ'' for glycerol as measured in the entire frequency regime from 10^{-6} Hz to 10^{12} Hz. The dynamic glass transition is seen over many decades in frequency with varying temperature. In addition in ϵ'' the excess wing and the boson peak show up (see Chap. 5) [38]

measurements the device can be used with a cryostat such that the total sample chamber inside the oversized cavity is composed of fused silica which is nearly transparent in this spectral range. By that serious problems can be circumvented by heat conduction in waveguide systems or by standing waves between the windows of the cryostat in quasioptical systems. The main disadvantage of oversized cavities is the loss of phase information. Thus only indirect measurements of the real part ϵ' of the complex dielectric function are possible. Details are discussed in [37].

For frequencies above 10^{12} Hz Fourier-Transform spectrometers are employed which are also available for the far infrared (e.g., Bruker 113v and 66v/s). If only the transmission and the reflection spectra are measured the complex permittivity can be determined by a Kramers-Kronig transformation. Using dispersive Fourier-Transform spectroscopy the complex dielectric function is measured directly [33, 34].

The complex dielectric spectrum of the glass-forming liquid glycerol as measured from 10^{-6} Hz to 10^{13} Hz is shown in Fig. 2.14. It is nicely demonstrated how well the different measurement systems can be combined.

2.5 Conclusions

With modern equipment it is possible to cover the entire regime from 10^{-6} Hz to 10^{12} Hz without any gap. For that different measurement systems based on different physical principles have to be combined: Fourier correlation analyzers (10^{-6} Hz to 10^7 Hz), impedance analyzers (10 Hz to 10^6 Hz), coaxial line reflectometers (10^6 Hz to 10^9 Hz), network analyzers (10^7 Hz to 10^{10} Hz), quasi-optical interferometers (10^{10} Hz to 10^{12} Hz), and Fourier spectrometers (10^{11} Hz to IR). In the time domain measurement systems between 10^{-6} Hz and 10^{10} Hz can be employed but with reduced accuracy.

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List of Abbreviations and Symbols

c	Speed of light
C_0	Vacuum capacitance
$E^*(\omega)$	Complex sinusoidal electric field strength
$J^*(\omega)$	Complex sinusoidal current density
$M^*(\omega) = M'(\omega) + iM''(\omega)$	Complex electrical modulus
$n^*(\omega) = n'(\omega) - in''(\omega)$	Complex index of refraction
R	Resistor
$r^*(x)$	Complex reflection at position x
$t^*(x)$	Complex transmission at position x
$U^*(\omega)$	Complex voltage
$Z^*(\omega)$	Complex impedance
α	Damping constant
β	Propagation constant
$\beta_{\epsilon}, \beta_M$	KWW stretching parameter for the dielectric permittivity and for the electric modulus
$\tan \delta = \frac{\epsilon''}{\epsilon'}$	Tangents of the phase angle between $\epsilon'(\omega)$ and $\epsilon''(\omega)$
$\epsilon(t)$	Time dependent dielectric function
ϵ_0	Permittivity of free space ($\epsilon_0 = 8.854 \times 10^{-12} \text{ As V}^{-1} \text{ m}^{-1}$)
$\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega)$	Complex dielectric permittivity or function ⁴
$\mu^*(\omega) = \mu'(\omega) - i\mu''(\omega)$	Complex magnetic permeability
λ	Wavelength in vacuum
$\bar{\lambda}$	Wavenumber
ν	Frequency
σ_{dc}	Direct current conductivity
$\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega)$	Complex conductivity
τ_{ϵ}, τ_M	Relaxation time of the KWW-function for the dielectric permittivity and for the electric modulus
$\phi(\omega)$	Phase shift
ω	Angular frequency

⁴ The complex index of refraction and the complex dielectric permittivity are related by Maxwell's relation $\epsilon^* = (n^*)^2$

$$n' = \left(\frac{\epsilon'}{2} \right)^{\frac{1}{2}} \left\{ \left[1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]^{\frac{1}{2}} + 1 \right\}^{\frac{1}{2}}$$

$$n'' = \left(\frac{\epsilon'}{2} \right)^{\frac{1}{2}} \left\{ \left[1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]^{\frac{1}{2}} - 1 \right\}^{\frac{1}{2}}$$

$$\epsilon' = n'^2 - n''^2$$

$$\epsilon'' = 2n'n''$$

a.c.	Alternating current
d.c.	Direct current
BWO	Backward-wave oscillator
FRA	Frequency response analyzer
KWW	Kohlrausch/Williams/Watts
RF	Radio frequency
TDR	Time domain reflectometer
VAPG	Variable amplitude-phase generator

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